

HALE COUNTY AP, TX, USA

WMO: 722174

Lat: 34.167N

Lon: 101.717W

Elev: 1028

StdP: 89.57

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 23097

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.6	-21.0	0.7	6.4	-17.4	0.9	3.3	14.6	13.5	12.8	13.1	3.2	10	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	36.2	18.0	35.1	18.9	33.8	19.1	23.0	29.0	21.9	29.0	21.3	28.3	5.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.2	25.3	20.2	16.9	24.0	19.4	16.1	23.7	74.1	28.6	69.7	28.8	66.9	27.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.0	9.4	-13.7	38.7	2.5	1.9	-15.6	40.1	-17.0	41.2	-18.4	42.3	-20.2	43.7		
(5)			-14.6	24.0	2.2	1.9	-16.2	25.3	-17.5	26.4	-18.7	27.4	-20.3	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	4.2	6.2	10.8	14.7	19.3	24.7	25.4	25.0	21.2	15.1	9.3	4.6		
	DBStd	8.86	5.11	5.63	5.10	4.72	4.66	2.88	2.38	2.48	3.25	4.71	4.91	5.08		
	HDD10.0	645	189	129	51	13	2	0	0	0	0	13	71	177		
	HDD18.3	2009	438	341	236	124	44	1	0	0	11	119	271	425		
	CDD10.0	2505	9	22	77	156	291	441	477	466	336	171	49	10		
	CDD18.3	828	0	0	3	16	76	192	219	207	97	18	1	0		
	CDH23.3	9689	4	26	119	372	1036	2201	2406	2197	996	283	47	3		
	CDH26.7	4504	1	3	22	117	469	1125	1198	1097	393	71	7	0		
(14) Wind	WSAvg	4.5	4.5	4.8	5.1	5.6	5.0	4.9	3.7	3.5	3.6	4.1	4.3	4.4		
(15) Precipitation	PrecAvg	509	15	16	24	33	73	83	64	58	57	44	23	17		
	PrecMax	867	67	56	76	144	217	255	298	204	137	165	142	65		
	PrecMin	226	0	0	0	0	4	8	0	0	0	0	0	0		
	PrecStd	141	16	14	21	31	49	55	57	44	41	46	25	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.5	28.9	32.5	36.2	38.0	37.6	37.1	35.0	32.1	27.7	23.0		
		MCWB	8.5	10.6	11.6	13.1	15.2	16.8	18.4	18.8	18.5	15.7	11.8	9.2		
	2%	DB	20.9	23.5	26.5	29.9	34.8	36.3	35.9	35.3	32.7	28.4	24.6	20.0		
		MCWB	8.7	9.7	11.8	12.3	15.1	17.4	19.1	19.8	18.4	15.7	11.2	8.8		
	5%	DB	17.1	20.7	24.1	27.6	32.2	35.0	34.8	34.7	31.9	26.5	22.0	16.4		
		MCWB	6.8	8.9	11.0	12.2	15.0	18.9	19.9	20.0	18.4	15.6	10.5	7.0		
	10%	DB	13.9	16.9	21.3	25.2	28.9	32.7	32.5	32.4	28.7	24.4	18.8	13.6		
		MCWB	5.8	6.9	10.4	12.4	15.3	18.8	20.2	20.1	18.0	14.1	9.6	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.5	16.0	17.9	20.7	23.9	26.4	23.2	21.9	20.5	15.0	12.1		
		MCDB	17.4	20.4	22.3	23.8	26.6	28.9	30.9	31.2	28.4	25.4	20.4	15.5		
	2%	WB	9.3	10.9	14.3	16.3	19.5	22.4	23.9	22.2	21.0	19.0	13.4	10.0		
		MCDB	18.0	20.8	20.1	22.7	25.6	29.1	28.5	30.2	26.8	23.9	18.8	16.3		
	5%	WB	7.6	9.4	13.0	15.1	18.6	21.5	22.2	21.6	20.3	17.8	12.1	8.5		
		MCDB	16.1	19.1	19.8	20.9	25.0	28.7	28.1	29.4	26.2	22.5	18.5	16.0		
	10%	WB	6.1	7.7	11.5	13.8	17.6	20.7	21.3	20.8	19.6	16.3	10.7	6.7		
		MCDB	13.8	16.2	19.0	20.9	24.4	27.9	27.8	28.5	25.6	21.8	18.1	13.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.5	16.2	16.1	14.7	14.3	13.2	13.6	13.3	14.7	15.4	14.0		
		MCDBR	20.4	21.0	20.3	20.1	18.9	17.0	15.7	15.7	17.0	17.8	20.0	18.8		
	5% WB	MCWBR	10.9	10.6	9.5	8.5	7.2	5.5	4.8	5.1	6.0	7.7	9.3	10.1		
		MCWBR	19.0	19.7	16.3	15.4	14.5	13.9	12.3	13.5	12.6	13.4	16.3	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.268	0.295	0.327	0.340	0.380	0.384	0.373	0.343	0.311	0.282	0.264		
		taud	2.587	2.567	2.499	2.405	2.433	2.364	2.400	2.393	2.454	2.523	2.536	2.565		
	Ebn at Noon		974	994	984	959	943	901	895	902	918	929	932	945		
		Edh at Noon	76	87	101	117	115	123	118	117	105	90	80	74		
(44) All-Sky Solar Radiation	RadAvg		3.29	4.07	5.32	6.51	7.07	7.52	7.23	6.58	5.51	4.33	3.43	2.83		
	RadStd		0.30	0.38	0.38	0.45	0.57	0.44	0.32	0.30	0.35	0.48	0.34	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (13 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	N/A	

Nomenclature: See separate page